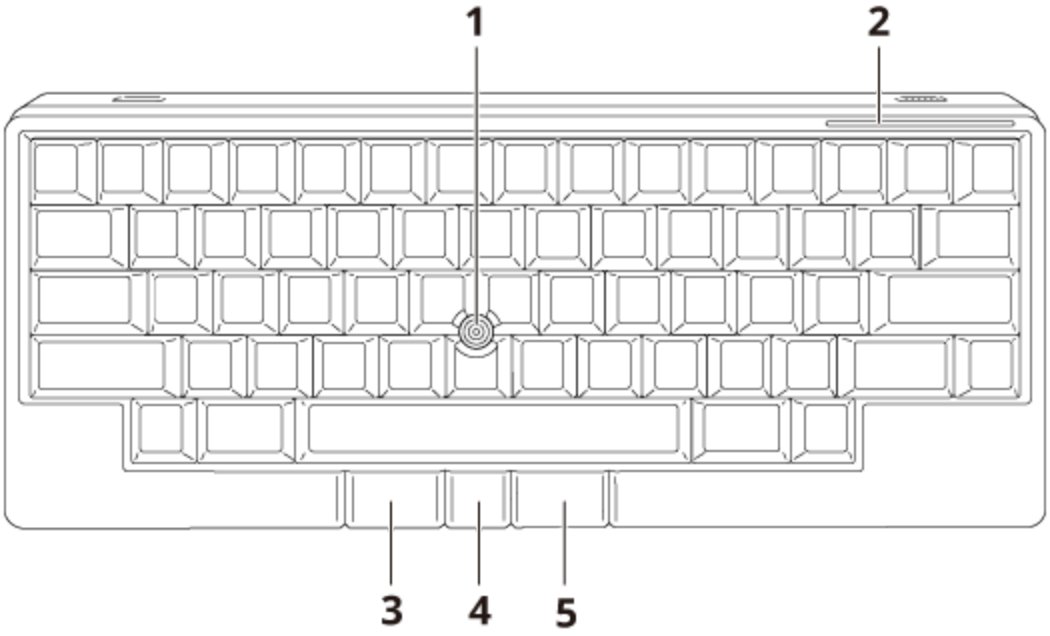
Happy Hacking Keyboard Studio

(US Layout) — Complete User Manual

Manufactured by PFU Limited
Issue Date: September 2024 | Document No. P3PX-E061-04ENZ2



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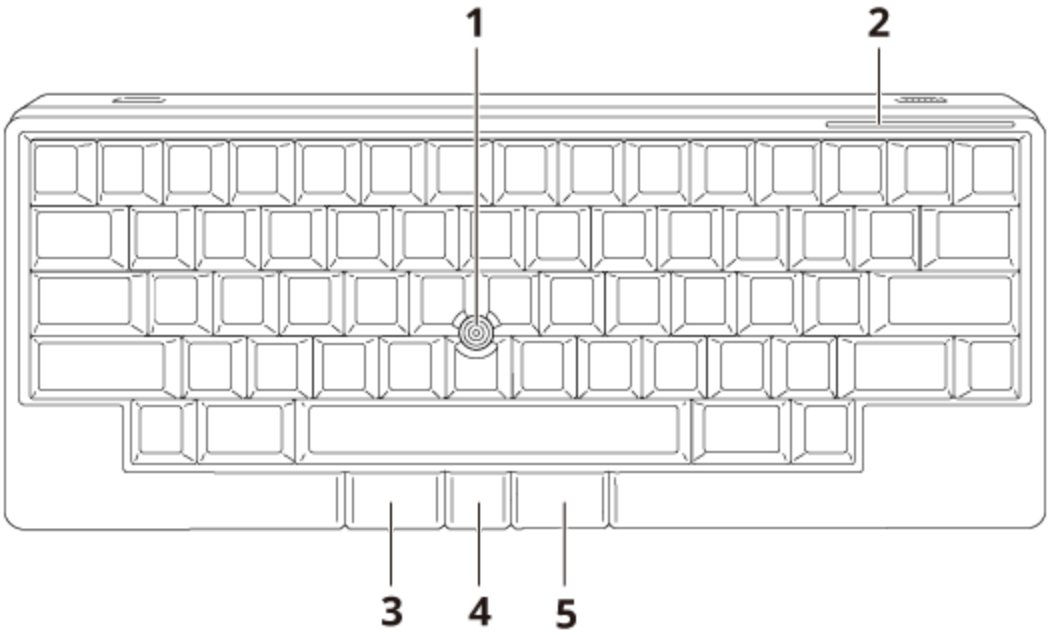
1. Parts and Functions

Front



No.	Name	Description
1	Pointing stick	Push this stick with your finger in any direction to move the cursor on the screen. Hold down the mouse key center button and control the pointing stick to scroll the screen.
2	LED indicator	Located at the upper right. Indicates the keyboard status with its color and blinking/lighting pattern.
3	Mouse key left button	Functions as the left button on an ordinary mouse by default.
4	Mouse key center button	Hold down this button and control the pointing stick to scroll. Functions as the [Fn2] key by default and can change pointing stick and gesture pad movements when used with other keys.
5	Mouse key right button	Functions as the right button on an ordinary mouse by default.

Back

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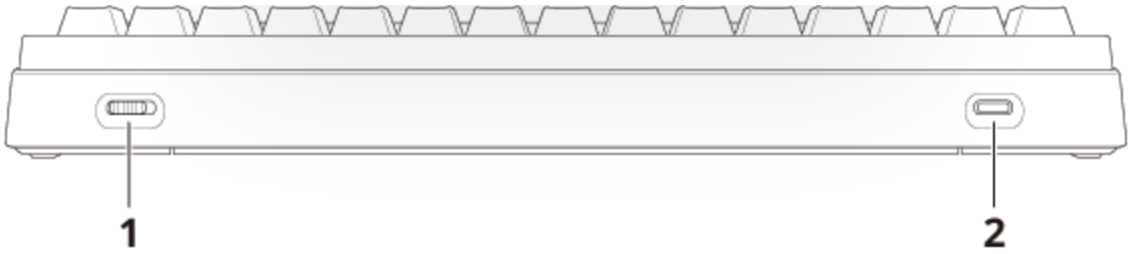
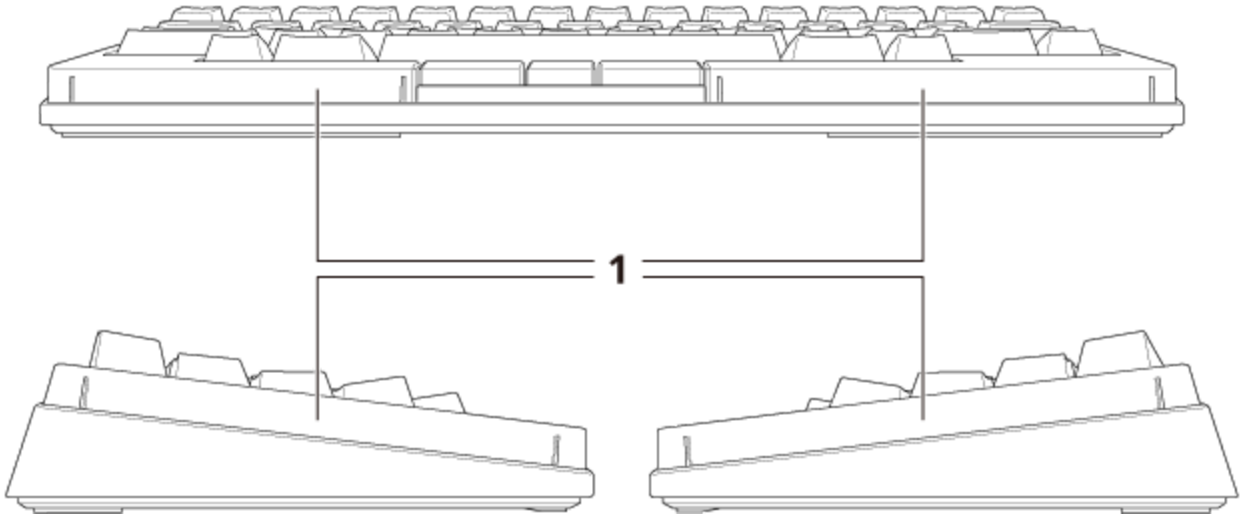


No.	Name	Description
1	Power switch	Turns on/off the keyboard. Slide right to turn on (green indicator visible).
2	USB Type-C connector	Used with the provided USB Type-C cable to connect the keyboard to a device via USB. This connector has no charging function for rechargeable batteries.

Side



No.	Name	Description
1	Gesture pads	Four pads located on the sides of the keyboard. Switch windows, scroll, and perform other operations intuitively by sliding the gesture pads with your fingers.

Bottom

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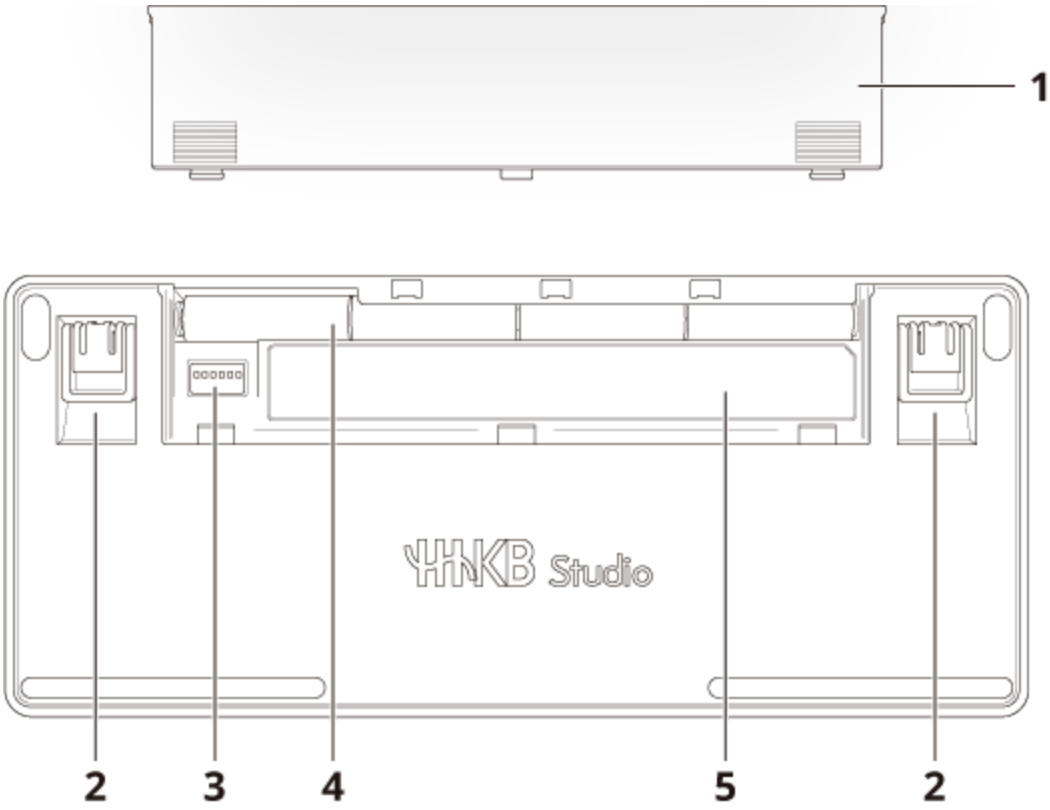

No.	Name	Description
1	Battery cover	Open this cover to insert/remove the batteries and access the DIP switches. Access information is printed on the reverse side.
2	Tilt stand	Used to adjust the height of the keyboard at two levels.
3	DIP switch	Turn the switch Up (ON)/Down to change the keyboard settings.
4	Battery box	To use the keyboard with a wireless connection, insert four AA batteries.
5	Product label	A label displaying the serial number (SER.NO.) and other information.

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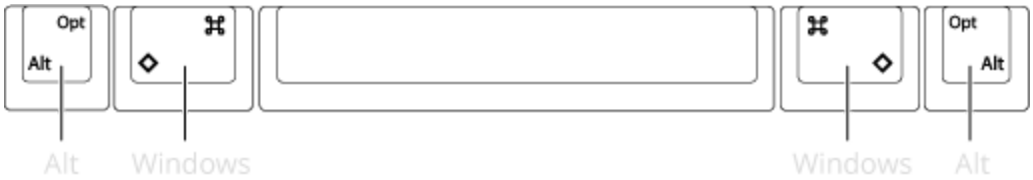
2. Special Key Assignment

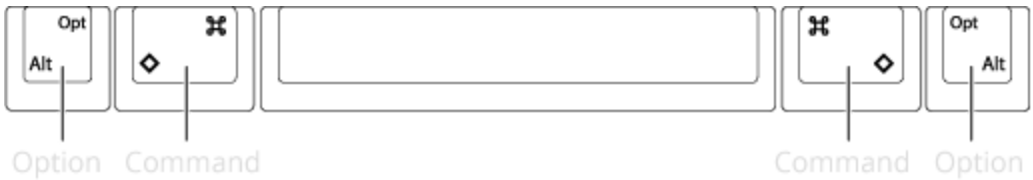
Special key assignment varies depending on the profile (key layout) in use. A profile for each OS is saved as default. Profile1 (Windows) is applied automatically when the keyboard is first connected.

To use the keyboard on macOS/iOS/iPadOS, switch to Profile2 (macOS profile).

Profile1 — Windows Key Layout



Profile2 — macOS Key Layout



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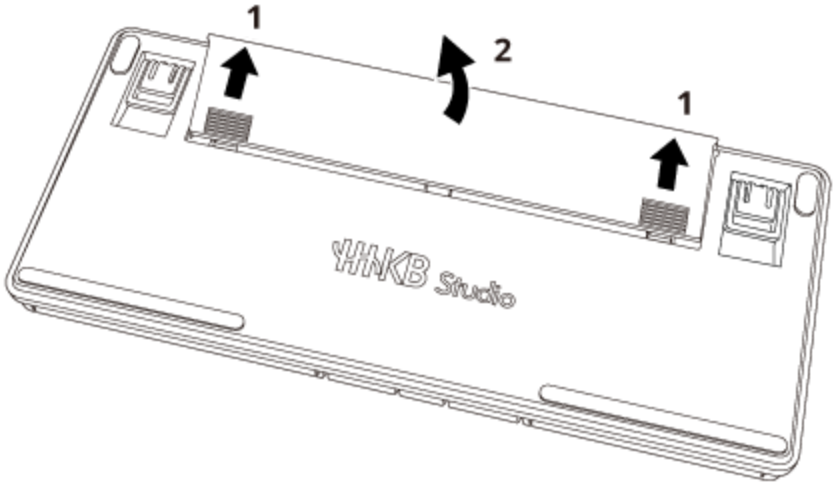
3. Power

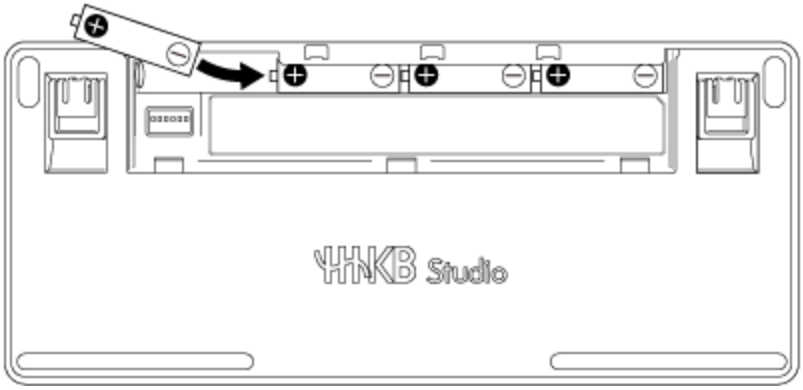
Using Batteries for Power Supply

To use the keyboard wirelessly, insert four AA batteries. When replacing batteries, make sure the power is off.



Procedure:

1. Slide the cover on the base of the keyboard and lift it up to remove it.
2. When lifting or moving the keyboard, hold it with both hands.
3. Insert four AA batteries, negative side first, into the battery box. Check that positive and negative terminals are placed correctly.
4. Attach the cover back. Close the cover firmly.



Using a USB Cable

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By connecting a USB cable to HHKB Studio, you can supply power via USB without using batteries. Connect HHKB Studio to a device with the provided USB Type-C cable. This does not consume battery power (except for self-discharge). Remove the batteries if you will not use them for a long time.

Turning the Power ON/OFF

Turning ON: Slide the power switch on the back of the keyboard to the right (it will turn green).

Turning OFF: Slide the power switch on the back of the keyboard to the left.

Power Saving Mode: If HHKB Studio is left unused for 30 minutes, it will automatically enter power saving mode and disconnect from Bluetooth. To reconnect, press the [Return] key. Power saving mode will not activate when connected via USB. You can change the power saving setting using DIP switch SW6.

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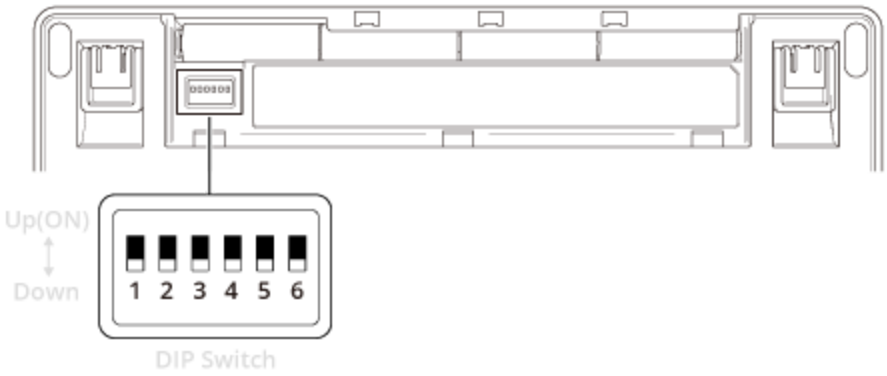
4. DIP Switch Settings

You can use the DIP switches on the keyboard to quickly change settings for the mouse keys, gesture pads, pointing stick, scrolling direction, [Delete] key, and power saving.

> **Important:** Be sure to turn off the power before setting any of the DIP switches. Use a flathead screwdriver to change the switches. All switches are set to [Down] as the factory default.



DIP Switch	SW1	SW2	SW3	SW4	SW5	SW6
Description	Mouse key	Gesture pad	Pointing stick	Scrolling direction	[Delete] key	Power saving
Up (ON)	Disabled	Disabled	Disabled	Reverse scrolling	Back Space	Disabled
Down	Enabled	Enabled	Enabled	Natural scrolling	Delete	Enabled

SW4 Notes: Reverse scrolling = opposite direction to the pointing stick push. Natural scrolling = same direction as the pointing stick push.

SW5 Note: For a macOS profile, SW5 is disabled and [Delete] always works as the Delete key.

SW6 Notes: When enabled (Down), the keyboard enters power saving mode after 30 minutes of inactivity. Press [Return] to reconnect. Power saving mode does not activate when connected via USB regardless of this setting. Requires firmware B0.07 or later.

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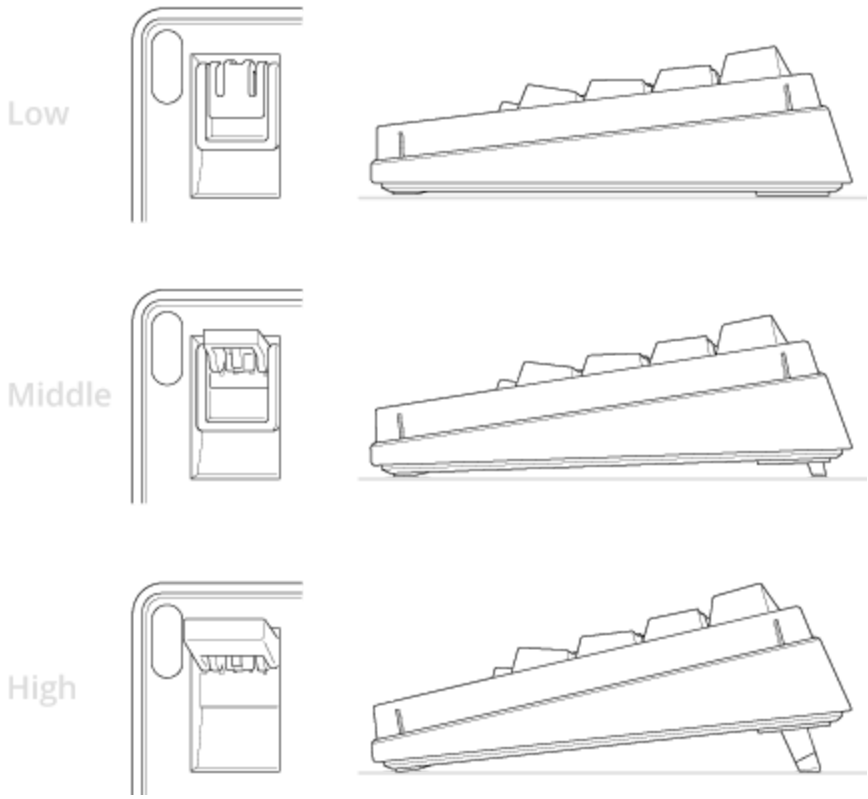
5. Adjusting Keyboard Height

A tilt mechanism on the bottom of the keyboard allows you to adjust the height at two levels. Raise one of the tilt stands until it locks in place. Set the tilt stands to the same height on the right and left sides.



To store the tilt stands, push back the long tilt stand first and then push back the short tilt stand. Pushing back both at the same time also works.

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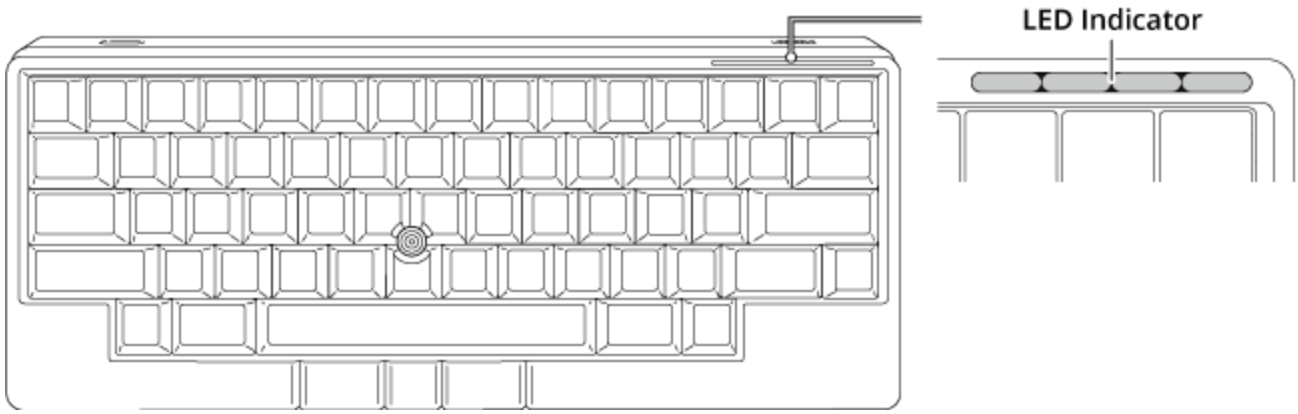
6. Status of the LED Indicator

The LED indicator on the upper right of HHKB Studio indicates the status of the keyboard. Various statuses are displayed by combining four LED indicator lights.



After Power On / Checking Profile & Device

Order	Status	Description
1	All four lights light up once in white	Indicates the keyboard is turned on.
2	Light for profile number lights up in white	Indicates which profile (1–4) is applied.
3a	Light for device lights up in blue (Bluetooth)	Shows which Bluetooth device (1–4) is connected.
3b	All four lights light up in blue (USB)	Indicates USB connection.

During Use

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Status	Description
All lights off	**Connected mode (Ready):** Keyboard is connected and ready. **Power saving mode:** Activated after 30 min inactivity (battery), 10 min in pairing mode, or when disconnected. Press [Return] to resume.
Leftmost light blinks in orange	Once every 30 seconds: battery power is low. Twice every 15 seconds: replace batteries immediately.

When Establishing/Switching a Connection

Status	Description
Pairing Standby Lights sweep side to side	**Pairing standby mode.** Press [Fn] + [Control] + [1]–[4] to enter pairing mode.
Pairing Mode Light blinks quickly in blue (5×/sec)	**Pairing mode.** Keyboard is ready to connect via Bluetooth.
BT Waiting Light lights up blue, then turns off	**Waiting for Bluetooth device** reconnection.
All four lights blue, then turn off	**Waiting for USB connection.**

When Switching Profiles

Status	Description
Profile Switch Light blinks in white (1×/sec)	**Profile switching mode.** The light for the current profile blinks.
Profile Switched Light lights up white, then turns off	**Profile switched** to the selected number.

When Adjusting Input Devices

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Status	Description
Blue/white lights per speed level	**Pointing stick cursor speed changed.** Number and color of lights vary by level (1–4 rough, 1–28 fine).
PS On All four white	Pointing stick temporarily **enabled**.
PS Off Lights at both ends white	Pointing stick temporarily **disabled**.
GP Sens 1–4 white lights from left	**Gesture pad sensitivity changed.** Low=1, Medium=2, High=3, Highest=4 lights.
GP On/Off Individual lights on/off per pad	**Gesture pad enable/disable status.** Left to right: left side, front left, front right, right side. Lit = enabled, off = disabled.

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7. Profiles

You can customize the key layout and save up to four layout profiles to number keys [1] to [4]. HHKB Studio has four layers for each profile. Use the Keymap Tool software to customize layouts and create your own profiles.

How to Switch Between Profiles

1. Hold down [Fn] and press [C]. The keyboard enters profile switching mode — the LED light for the current profile blinks in white.
2. Press a number key [1]–[4] to switch to that profile.
3. The LED light for the selected profile lights up in white, then turns off.

How to Check the Profile in Use

Hold down [Fn] and press [C]. The LED light for the current profile blinks in white. Press [Fn] + [X] to exit without changing.

Factory Default Profiles

Profile	OS	Description
Profile1	Windows	Key layout optimized for Windows
Profile2	macOS	Key layout optimized for macOS/iOS/iPadOS
Profile3	(Empty)	Available for customization
Profile4	(Empty)	Available for customization

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8. Bluetooth Connection

Pairing information for up to four Bluetooth devices can be registered to number keys [1]–[4]. You can quickly switch between devices using shortcut keys.

> **Hint:** If your device doesn't support Bluetooth, use a commercially available Bluetooth USB adapter or the provided USB Type-C cable.

Connecting (Pairing) for the First Time

1. **Turn on** the keyboard by sliding the power switch to the right.
2. **Enter pairing mode.** When no pairing info exists, the keyboard enters pairing mode automatically. The leftmost LED blinks quickly in blue.
3. **Pair with your device** (see OS-specific instructions below).

Windows 11

a. Open Settings → Bluetooth & devices → Add device → Bluetooth
b. Select "HHKB-Studio1" from the list
c. A several-digit number for pairing is displayed — enter it on the keyboard and press [Return]
d. When "Connected" appears for HHKB-Studio1, the connection is established

Windows 10

a. Open Windows Settings → Devices → Bluetooth & other devices → Add Bluetooth or other device → Bluetooth
b. Select "HHKB-Studio1" from the list
c. Enter the pairing number on the keyboard and press [Return]
d. When "Connected" appears, the connection is established

macOS

a. Open System Preferences/Settings → Bluetooth
b. Select "HHKB-Studio1" and enter the pairing number on the keyboard, press [Return]
c. If [Connect] button is available, click it
d. **Switch to Profile2 (macOS):** Press [Fn] + [C], then press [2]

iOS / iPadOS

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a. Open Settings → Bluetooth
b. Select "HHKB-Studio1" and enter the pairing number, press [Return]
c. **Switch to Profile2:** Press [Fn] + [C], then press [2]

Android

a. Open Settings → Connected devices → Pair new device
b. Select "HHKB-Studio1" and enter the pairing number, press [Return]

Registering an Additional Bluetooth Device

1. Turn on the keyboard.
2. Press [Fn] + [Q] to enter pairing standby mode (LED lights sweep side to side).
3. Press [Fn] + [Control] + [1]–[4] to assign the pairing to a specific number key and enter pairing mode.
4. Follow the OS-specific pairing steps above.
5. Switch to the appropriate profile for your OS if needed.

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9. USB Connection

Connect HHKB Studio to a device with the provided USB Type-C cable for a wired connection. This does not consume battery power (except for self-discharge).

1. Turn on the power by sliding the power switch to the right.
2. **If connected for the first time or was last connected via USB:** The keyboard automatically switches to USB connection. All four LED lights light up in blue, then turn off.
3. **If currently connected or was last connected via Bluetooth:** Press [Fn] + [Control] + [0] to switch to USB connection. All four LED lights will light up in blue, then turn off.

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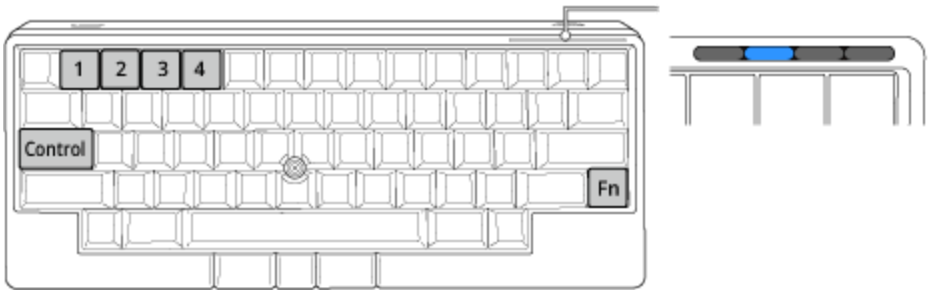
10. Switching & Deleting Devices

Switching Between Devices

When turned on, HHKB Studio automatically connects to the last-connected device.

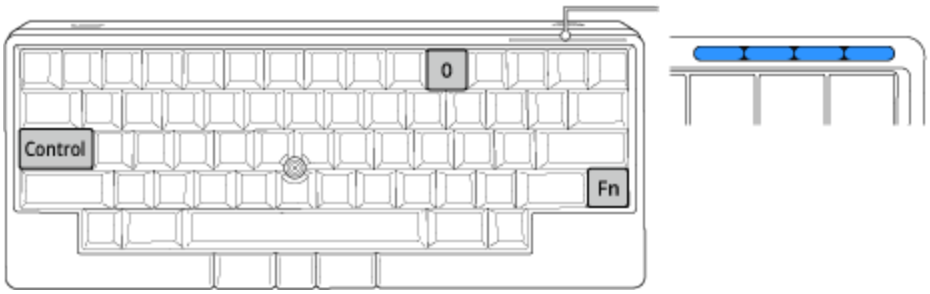


Switch to Bluetooth device

Press [Fn] + [Control] + [1]–[4] (the number key where the pairing info is registered). The corresponding LED light lights up in blue during connection, then turns off.

Switch to USB connection

Press [Fn] + [Control] + [0]. All four LED lights light up in blue, then turn off.



Check connected device

Press [Fn] + [Control] + [0]. Bluetooth: the light for the device number lights up in blue. USB: all four lights light up in blue.

Deleting Pairing Information

Deleting Individually

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1. Press [Fn] + [Q] to enter pairing standby mode (lights sweep side to side).
2. Press [Fn] + [Control], then hold [Delete] and press the number key [1]–[4] for the device to delete.
3. The LED turns off and the keyboard disconnects.
4. Also delete the device from your OS Bluetooth settings.

Press [Return] to reconnect after deletion.

Deleting All at Once

1. Press [Fn] + [Q] to enter pairing standby mode.
2. Press [Fn] + [Control], then hold [Delete] and press [0].
3. All pairing information is deleted.
4. Also delete from your OS Bluetooth settings.

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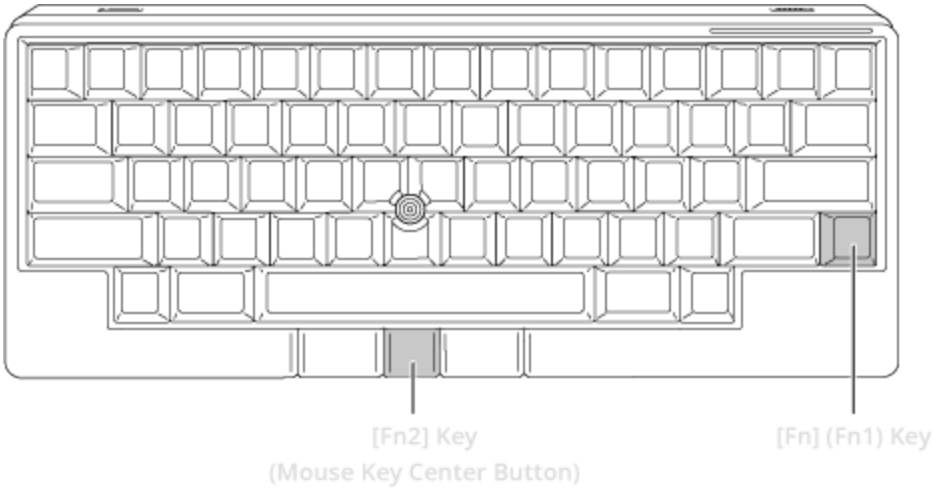
11. Fn (Function) Key & Layers

HHKB Studio has four layers for each profile, activated by the function keys [Fn] (Fn1), [Fn2], and [Fn3].

Function Key Positions



Key	Default Position
[Fn] (Fn1)	Right side of the bottom row
[Fn2]	Mouse key center button
[Fn3]	Not assigned — customizable via Keymap Tool

Layer Types

Layer	Activation	Default Function
Standard	No Fn key pressed	Normal key input
Fn1 layer	Hold [Fn]	F1–F12, PgUp/PgDn, Home/End, arrow keys, Caps Lock, volume controls, and connection key combinations
Fn2 layer	Hold [Fn2] (mouse center)	Cursor speed adjustment, gesture pad sensitivity
Fn3 layer	Hold [Fn3]	Fully customizable via Keymap Tool

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12. Fn Key Input Reference Table

To input characters printed on the front side of the key tops, hold down the [Fn] key and press the corresponding key.

Key	Front Print	Meaning	Win	Mac
[1]–[=]	F1–F12	Function keys F1 through F12	✓	✓
(key)	Ins	Insert	✓	—
(key)	Del	Delete	✓	✓
(key)	Clear	Clear (keypad)	—	✓
(key)	PS/SRq	Print Screen / System Request	✓	—
(key)	ScrLk	Scroll Lock	✓	—
(key)	Pus/Brk	Pause / Break	✓	—
(key)	Home	Home	✓	✓
(key)	PgUp	Page Up	✓	✓
(key)	End	End	✓	✓
(key)	PgDn	Page Down	✓	✓
(key)	Caps	Caps Lock	✓	✓
Arrow keys	←↑→↓	Cursor keys	✓	✓
(key)	VolDn	Volume Down	✓	✓
(key)	VolUp	Volume Up	✓	✓
(key)	Mute	Mute	✓	✓
(key)	Eject	Eject	—	✓
(key)	Power	Power (turns off connected device)	—	✓

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13. Key Combinations Quick Reference

Operation	Key Combination	LED Status
Enter pairing mode (first time)	[Fn] + [Q]	Leftmost light blinks blue
Register additional pairing	[Fn] + [Q] → [Fn] + [Control] + [1]–[4]	Corresponding light blinks blue
Switch to Bluetooth device	[Fn] + [Control] + [1]–[4]	Corresponding light lights blue
Switch to USB	[Fn] + [Control] + [0]	All four lights blue
Check connection status	[Fn] + [Control] + [0]	Blue lights indicate type
Delete individual pairing	[Fn] + [Q] → [Fn] + [Control] + [Delete] + [1]–[4]	Lights turn off
Delete all pairing	[Fn] + [Q] → [Fn] + [Control] + [Delete] + [0]	Lights turn off
Switch profile	[Fn] + [C] → [1]–[4]	Corresponding light white
Check current profile	[Fn] + [C]	Current profile light blinks white
Exit pairing standby / profile mode	[Fn] + [X]	—
Cursor speed (4 levels)	Mouse center + [1]–[4]	Blue/white lights
Cursor speed fine adjust	Mouse center + [↑] / [↓]	Blue/white lights
Gesture pad sensitivity	Mouse center + [6]–[9]	1–4 white lights

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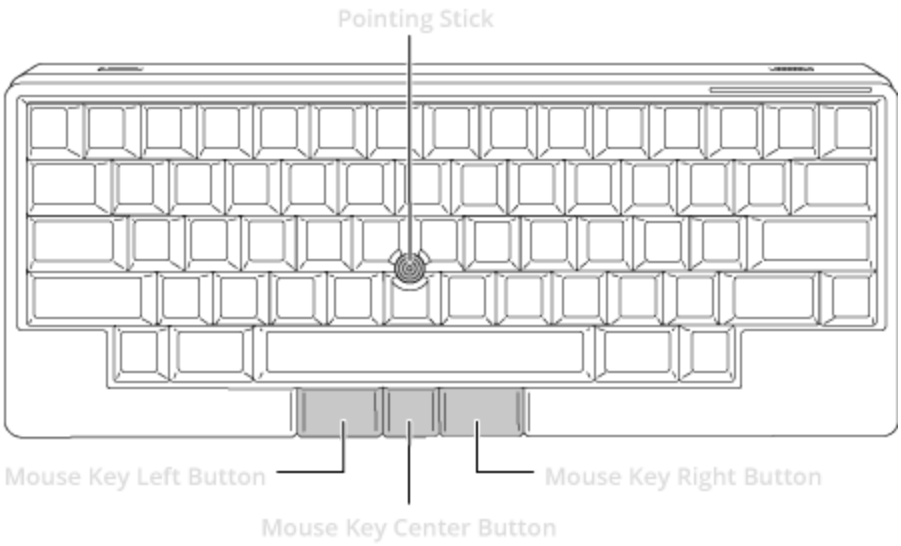
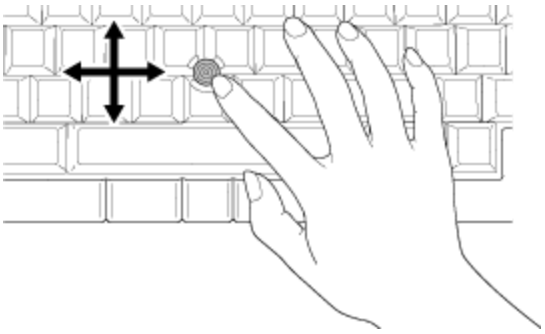
14. Using the Pointing Stick & Mouse Keys

How to Operate



Moving the Cursor

Push the pointing stick with your fingertip in any direction to move the cursor.



Scrolling

Hold down the mouse key center button and push the pointing stick in the desired direction.

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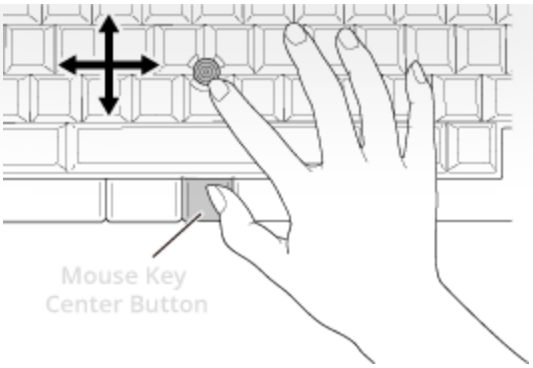

Pointing Stick Button

Press the pointing stick straight down to perform a mouse middle-click.

Mouse Key Operations

Button	Action	Result
Left button	Press once	Left-click
Left button	Press twice quickly	Double-click
Left button	Hold and move pointing stick	Drag
Right button	Press once	Right-click
Center button	Hold	Scroll mode (move pointing stick to scroll)
Center button	Functions as [Fn2]	Activates Fn2 layer key combinations

> **Note:** The pointing stick cursor may drift or lag when first used after power on, after being enabled, when used with excessive force, after prolonged use, or during temperature changes. Wait a few seconds without touching the pointing stick, then try again, or restart the keyboard.

Changing the Cursor Speed

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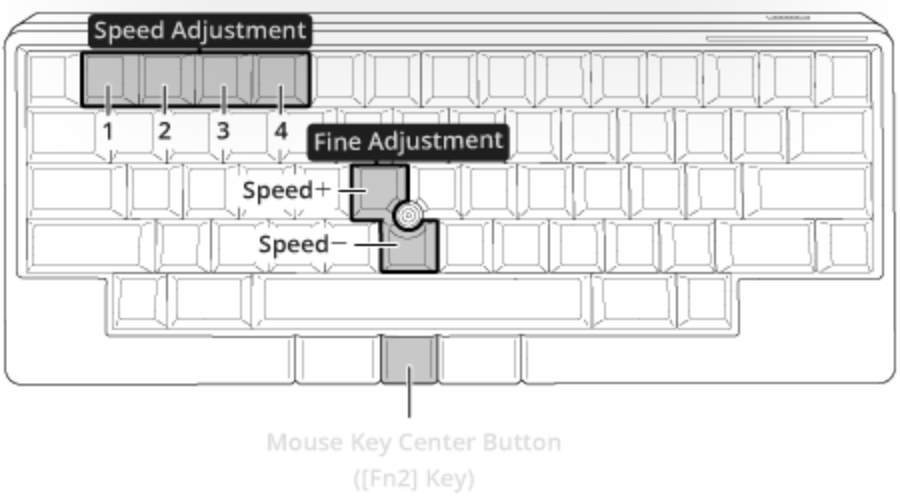

Default speed: Speed 1. Setting range: 4 levels (rough) or 28 levels (fine).

Key Combination	Speed	LED Status
Mouse center + [1]	Speed 1 (slow)	Leftmost light white
Mouse center + [2]	Speed 2	Left blue + second white
Mouse center + [3]	Speed 3	Left two blue + third white
Mouse center + [4]	Speed 4 (fast)	Left three blue + rightmost white
Mouse center + [↑]	Increase by 1 step	Slow Fine adjustment (28 levels) Fast
Mouse center + [↓]	Decrease by 1 step	

Changing the Scrolling Direction

Use DIP switch SW4 on the base of the keyboard:

SW4 Position	Scrolling Behavior
Up (ON)	**Reverse scrolling:** Opposite direction to the pointing stick push
Down (default)	**Natural scrolling:** Same direction as the pointing stick push

Enabling/Disabling

Claude is active in this tab group

DIP switches: SW1 controls mouse keys (Up=disabled, Down=enabled). SW3 controls pointing stick (Up=disabled, Down=enabled).

Key operation (temporary): Configurable via Keymap Tool. Settings are not saved — power cycling restores DIP switch settings.

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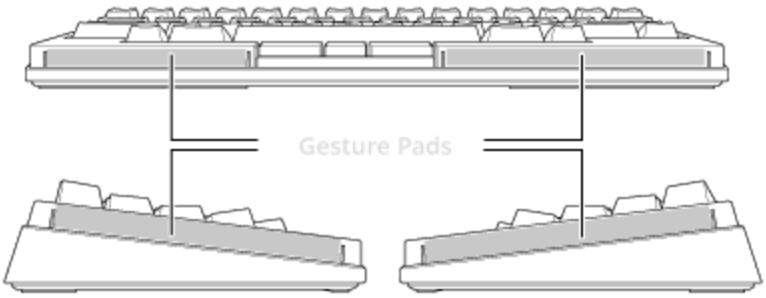
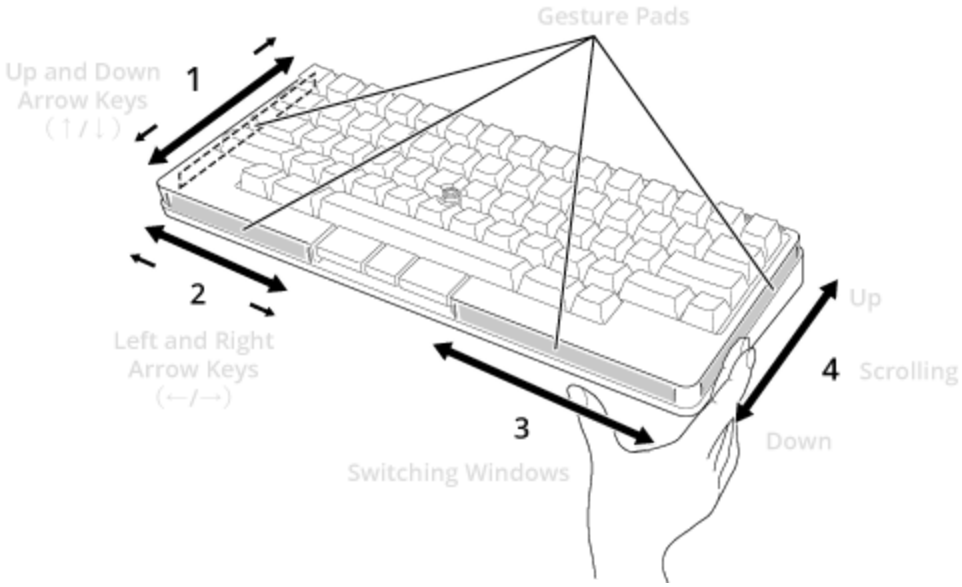
15. Using the Gesture Pads

The gesture pads are four touch-sensitive pads on the sides of the keyboard (two on the front, one on each side). Slide your finger on a pad to activate the assigned function.



> **Note:** When lifting or moving the keyboard while it is turned on, hold it with both hands to prevent unintended gesture pad activation. If using a palm rest, leave a gap between the palm rest and keyboard.

Default Gesture Pad Assignments



Position	Direction	Default Function
Front left	Slide left/right	Left/Right arrow keys
Front right	Slide left/right	Switch foremost window (Alt+Tab / Cmd+Tab)
Left side	Slide forward/backward	Up/Down arrow keys
Right side	Slide forwa	

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Adjusting the Sensitivity

Default: Medium. Setting range: 4 levels.

Key Combination	Sensitivity	LED Status
Mouse center + [6]	Low	1 white light (leftmost)
Mouse center + [7]	Medium	2 white lights from left
Mouse center + [8]	High	3 white lights from left
Mouse center + [9]	Highest	All 4 white lights

Enabling/Disabling

DIP switch: SW2 Up (ON) = all gesture pads disabled. SW2 Down = all enabled.

Key operation (temporary): Configurable via Keymap Tool. Each pad can be toggled individually. LED indicators show status (left to right: left side, front left, front right, right side — lit = enabled, off = disabled).

Keymap Tool: Each gesture pad can be enabled/disabled individually, and settings are saved to the keyboard.

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16. Changing the Keymap (Keymap Tool)

Use the dedicated Keymap Tool software to change the key layout, register shortcuts, and change gesture pad movements.

What You Can Customize

Category	Details
Applicable range	Keys, mouse keys, and gesture pad slide operations
Basic key functions	Standard keyboard keys, media keys, screen brightness, scroll/wheel operation
Device functions	Mouse clicks, enable/disable gesture pads & pointing stick, cursor speed, gesture sensitivity, profile switching, connection switching
Shortcut keys	Single key combination shortcuts assigned to a single key
Macro keys	Up to 32 sequential key operations assigned to a single key

Getting Started

1. Visit the **HHKB Studio web portal:** https://happyhackingkb.com/global/studio/portal/
2. Connect HHKB Studio to your device via USB.
3. Access Keymap Tool from the web portal using a compatible web browser (Chrome recommended).

> **Important:** Keymap Tool works only with a USB connection. An internet connection is required to access the web portal.

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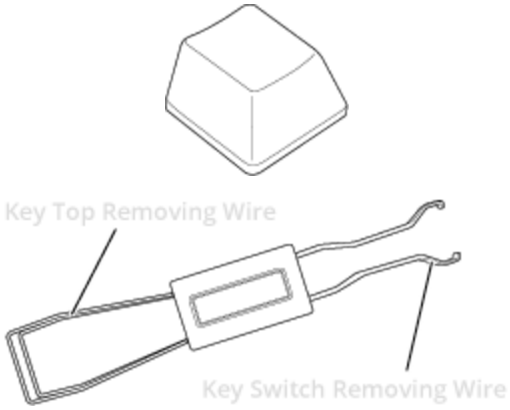
17. Replacing Parts

> **Warning:** Replace parts at your own risk. Parts may be damaged during replacement. Read the procedures carefully.

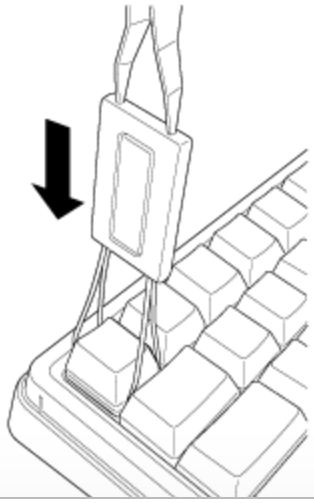
Replacing Key Tops

Key tops can be replaced with PFU optional parts or compatible third-party parts. Only parts designed for HHKB Studio are compatible (Professional series parts will **not** fit).



Procedure:

1. Prepare a new key top and a remover tool.
2. Turn off the keyboard.
3. Insert the remover tool under the key top and pull straight upward to remove it.



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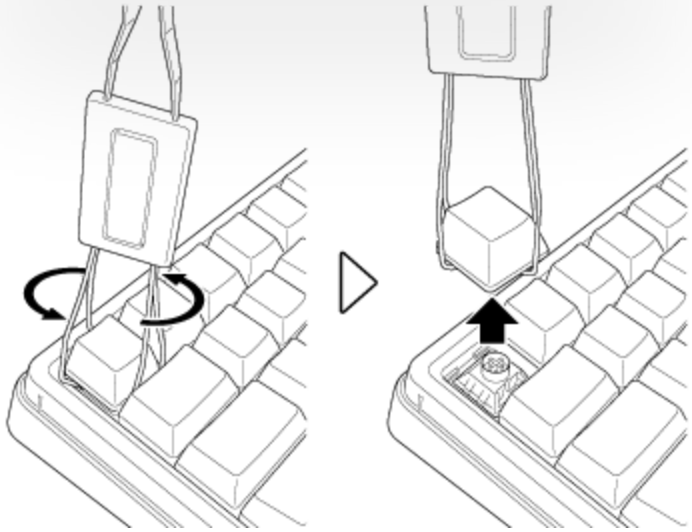
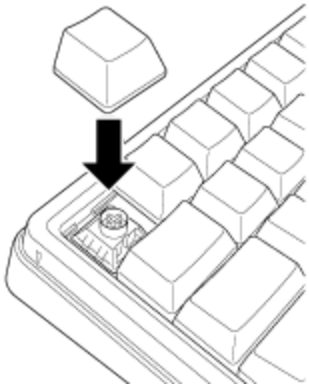
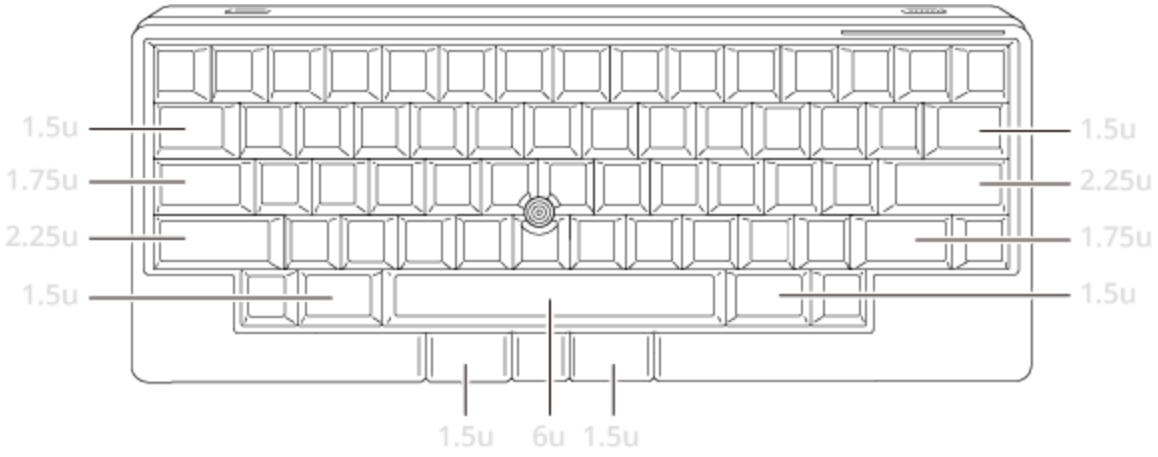


4. Place the new key top on the key switch and press down firmly until it clicks.



Key Top Size Reference



Replacing Key Switches

Key switches can be replaced with PFU optional parts or compatible third-party parts. Only HHKB Studio switches are compatible — Professional series are **not** replaceable.

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The procedure covers both standard key switches and mouse key switches. Follow the detailed instructions carefully to avoid damage.

Replacing the Pointing Stick Cap

The pointing stick cap can be replaced with the spare cap supplied with the keyboard.

1. Pull the pointing stick cap straight upward to remove it.
2. Align the hole on the new cap with the protrusion on the keyboard and push it firmly to the bottom.

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18. Keyboard Specifications

Item	Specification
Key specification	Mechanical / Step sculpture / Key pitch: 19.05 mm (0.75 in.)
Ambient temperature	5 to 35°C (41 to 95°F)
Ambient humidity	20 to 80% RH (non-condensing)
Dimensions	308 mm (W) × 132 mm (D) × 41 mm (H) / 12.13 × 5.20 × 1.61 in.
Weight	Approx. 840 g / 1.85 lb (without batteries)
Connection method	Bluetooth wireless; USB 2.0 Full Speed (Type-C)
Bluetooth standard	Bluetooth 5.0 (LE) Class 2
Bluetooth profile	HOGP (HID over GATT profile)
Radio frequency	2.4 GHz band, GFSK modulation
Wireless range	Approx. 10 m (no magnetic objects nearby)
Battery life	Approx. 3 months (with AA alkaline; varies by usage)
Required batteries	4× AA (alkaline, manganese, or NiMH rechargeable)

OS Compatibility

Windows, macOS, iOS, iPadOS, Android, visionOS. For Bluetooth, device must support Bluetooth 5.0 (LE) Class 2.

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19. Cleaning

When the keyboard is dirty, wipe it with a soft, dry cloth. For stubborn dirt, wipe with a cloth moistened with a small amount of mild detergent. **Do not use volatile liquids such as alcohol** — they may damage the product in quality and color.

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